

Math 39 - Precalculus with Data Analysis

Middlesex School Mathematics | Resource: Math 39

COURSE AT A GLANCE

Linear Modeling and Residuals

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can calculate and interpret average rate of change with appropriate units.
- I can construct a linear function from two data points, a table, a graph, or a context.
- I can fit a linear regression model to data using technology and interpret its parameters.
- I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.
- I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.
- I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.
- I can construct an exponential function from two data points, a table, or a context.
- I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.
- I can determine and interpret doubling time or half-life from an exponential model.
- I can solve exponential and logarithmic equations in modeling contexts.
- I can fit an exponential regression model using technology and evaluate its fit.
- I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can identify a power function and interpret the parameters in $y = kx^p$.
- I can construct a power function from two data points or a proportional relationship.
- I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
- I can recognize and model direct, inverse, and other power-variation relationships.
- I can solve equations involving integer or rational powers.
- I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
- I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.
- I can determine where a combination of functions is positive, zero, or undefined.
- I can determine polynomial end behavior from degree and leading coefficient.
- I can identify zeros and multiplicities and relate them to polynomial graph behavior.
- I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.
- I can construct a polynomial function from specified zeros, multiplicities, and a point.
- I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.
- I can build and optimize a polynomial model in a geometric or economic context.

Describing Data

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can distinguish categorical and quantitative data.
- I can construct and interpret frequency and relative-frequency tables.
- I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.
- I can identify misleading features in a data display.
- I can describe a distribution's modality, symmetry, skewness, and outliers.
- I can calculate and interpret mean, median, and mode and select an appropriate measure of center.
- I can calculate and interpret range, standard deviation, quartiles, and interquartile range.
- I can construct and interpret a five-number summary and boxplot.
- I can use percentiles and z-scores to locate and compare values within distributions.
- I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can organize categorical data in a contingency table or Venn diagram.
- I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.
- I can identify an experiment, outcomes, events, and a sample space.
- I can calculate theoretical probabilities when outcomes are equally likely.
- I can use complements to calculate probabilities.
- I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.
- I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.
- I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can calculate the expected value of a discrete probability situation.
- I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

No mapped textbook section

UNIT LEARNING OBJECTIVES

- I can linearize power or exponential data and translate between the linearized and original models.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Linear Modeling and Residuals

REQUIRED

Resource coverage: No mapped textbook section

M39-LIN-001

I can calculate and interpret average rate of change with appropriate units.

Resource: No mapped textbook section

SUPPORTING SKILLS

- Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- Interpret average rate of change with contextual units.
SK-FUNC-AROC-INTERPRET · first introduced in Math 21
- Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M39-LIN-002

I can construct a linear function from two data points, a table, a graph, or a context.

Resource: No mapped textbook section

SUPPORTING SKILLS

- Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12
- Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-LIN-003

I can fit a linear regression model to data using technology and interpret its parameters.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- I** Use technology to determine a least-squares linear regression model.
SK-REG-LINEAR-FIT

M39-LIN-004

I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Interpret the sign and magnitude of a correlation coefficient in context.
SK-REG-CORRELATION-INTERPRET

M39-LIN-005

I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Use residual patterns and contextual evidence to judge whether a model is appropriate.
SK-REG-MODEL-ASSESS
- I** Calculate an observed value minus its model-predicted value.
SK-REG-RESIDUAL-CALCULATE
- I** Construct and interpret a residual plot.
SK-REG-RESIDUAL-PLOT

M39-LIN-006

I can use a linear model to make and interpret predictions, including solving for either variable.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12

Exponential Modeling

REQUIRED

Resource coverage: No mapped textbook section

M39-EXP-001

I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

Resource: No mapped textbook section

SUPPORTING SKILLS

D Use constant differences or ratios to distinguish linear and exponential patterns.

SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31

D Compare functions shown in different representations.

SK-FUNC-COMPARE-REP · first introduced in Math 21

M39-EXP-002

I can construct an exponential function from two data points, a table, or a context.

Resource: No mapped textbook section

SUPPORTING SKILLS

D Identify initial amount, rate, and time interval in context.

SK-EXP-MODEL-CONTEXT · first introduced in Math 31

D Determine an exponential model from tabular values.

SK-EXP-MODEL-TABLE · first introduced in Math 31

D Solve for an exponential model using two points.

SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31

M39-EXP-003

I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

Resource: No mapped textbook section

SUPPORTING SKILLS

D Convert a percent increase or decrease to a growth or decay factor.

SK-EXP-PERCENT-FACTOR · first introduced in Math 31

D Explain a model parameter in the language of its context.

SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M39-EXP-004

I can determine and interpret doubling time or half-life from an exponential model.

Resource: No mapped textbook section

SUPPORTING SKILLS

A Identify initial amount, rate, and time interval in context.

SK-EXP-MODEL-CONTEXT · first introduced in Math 31

D Translate a doubling-time or half-life question into an exponential equation.

SK-LOG-DOUBLING-HALF · first introduced in Math 31

M39-EXP-005

I can solve exponential and logarithmic equations in modeling contexts.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- R** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- R** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG · first introduced in Math 31

M39-EXP-006

I can fit an exponential regression model using technology and evaluate its fit.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Interpret the sign and magnitude of a correlation coefficient in context.
SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- I** Use technology to determine an exponential regression model.
SK-REG-EXPONENTIAL-FIT
- D** Use residual patterns and contextual evidence to judge whether a model is appropriate.
SK-REG-MODEL-ASSESS · first introduced in Math 39

M39-EXP-007

I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- I** Select a function family whose behavior fits a table, graph, or context.
SK-MODEL-FAMILY-SELECT
- D** Use residual patterns and contextual evidence to judge whether a model is appropriate.
SK-REG-MODEL-ASSESS · first introduced in Math 39

Power Functions and Function Comparison

REQUIRED

Resource coverage: No mapped textbook section

M39-POW-001

I can identify a power function and interpret the parameters in $y = kx^p$.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Interpret the coefficient and exponent in a power function.
SK-POWER-PARAMETER-INTERPRET
- I** Represent and interpret a power-variation relationship.
SK-VAR-POWER

M39-POW-002

I can construct a power function from two data points or a proportional relationship.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I** Determine a power function from two points or a proportional relationship.
SK-POWER-CONSTRUCT

M39-POW-003

I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Determine power-function behavior as the input approaches zero or positive or negative infinity.
SK-POWER-END-BEHAVIOR
- I** Relate the coefficient and exponent of a power function to its graph shape.
SK-POWER-SHAPE
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M39-POW-004

I can recognize and model direct, inverse, and other power-variation relationships.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Represent and interpret a direct-variation relationship.
SK-VAR-DIRECT
- I** Represent and interpret an inverse-variation relationship.
SK-VAR-INVERSE
- D** Represent and interpret a power-variation relationship.
SK-VAR-POWER · first introduced in Math 39

M39-POW-005

I can solve equations involving integer or rational powers.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21

M39-POW-006

I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- I** Compare the long-run growth or decay of functions from different families.
SK-FUNC-DOMINANCE

M39-POW-007

I can fit a power regression model and interpret its parameters and correlation evidence.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D** Interpret the coefficient and exponent in a power function.
SK-POWER-PARAMETER-INTERPRET · first introduced in Math 39
- A** Interpret the sign and magnitude of a correlation coefficient in context.
SK-REG-CORRELATION-INTERPRET · first introduced in Math 39
- I** Use technology to determine a power regression model.
SK-REG-POWER-FIT

Function Combinations and Polynomials

REQUIRED

Resource coverage: No mapped textbook section

M39-POL-001

I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I** Add, subtract, multiply, and divide function outputs.
SK-FUNC-OPERATIONS

M39-POL-002

I can determine where a combination of functions is positive, zero, or undefined.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Determine the domain of a sum, difference, product, quotient, or composition.
SK-FUNC-DOMAIN-COMBINE
- D** Add, subtract, multiply, and divide function outputs.
SK-FUNC-OPERATIONS · first introduced in Math 39

M39-POL-003

I can determine polynomial end behavior from degree and leading coefficient.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M39-POL-004

I can identify zeros and multiplicities and relate them to polynomial graph behavior.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY
- D** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M39-POL-005

I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39
- D** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39

M39-POL-006

I can construct a polynomial function from specified zeros, multiplicities, and a point.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT · first introduced in Math 31
- A** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M39-POL-007

I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D** Infer polynomial end behavior from degree and leading coefficient.
SK-POLY-END-BEHAVIOR · first introduced in Math 39
- A** Relate zero multiplicity to local graph behavior.
SK-POLY-MULTIPLICITY · first introduced in Math 39
- A** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS · first introduced in Math 21

M39-POL-008

I can build and optimize a polynomial model in a geometric or economic context.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D** Interpret vertex and zeros in an application.
SK-QUAD-MODEL-FEATURES · first introduced in Math 21

Describing Data

REQUIRED

Resource coverage: No mapped textbook section

M39-STA-001

I can distinguish categorical and quantitative data.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Classify data as categorical or quantitative.
SK-STAT-DATA-TYPE

M39-STA-002

I can construct and interpret frequency and relative-frequency tables.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Construct and interpret a frequency table.
SK-STAT-FREQUENCY-TABLE
- I** Calculate and interpret relative frequency.
SK-STAT-RELATIVE-FREQUENCY

M39-STA-003

I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Construct and interpret bar charts and pie charts.
SK-STAT-CATEGORICAL-DISPLAY
- I** Construct and interpret a histogram for quantitative data.
SK-STAT-HISTOGRAM

M39-STA-004

I can identify misleading features in a data display.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Identify perceptual distortion, misleading scales, and other misleading features in data displays.
SK-STAT-GRAPH-CRITIQUE

M39-STA-005

I can describe a distribution's modality, symmetry, skewness, and outliers.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Describe a distribution by modality, symmetry, and skewness.
SK-STAT-DISTRIBUTION-SHAPE
- I** Identify and interpret unusual observations or outliers.
SK-STAT-OUTLIER-IDENTIFY

M39-STA-006

I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Select an appropriate measure of center based on distribution shape and context.
SK-STAT-CENTER-SELECT
- I** Calculate and interpret the arithmetic mean.
SK-STAT-MEAN
- I** Calculate and interpret the median.
SK-STAT-MEDIAN
- I** Identify and interpret the mode or modes.
SK-STAT-MODE

M39-STA-007

I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Determine quartiles and calculate and interpret the interquartile range.
SK-STAT-QUARTILES-IQR
- I** Calculate and interpret the range.
SK-STAT-RANGE
- I** Calculate and interpret standard deviation as a measure of spread.
SK-STAT-STANDARD-DEVIATION

M39-STA-008

I can construct and interpret a five-number summary and boxplot.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Construct and interpret a boxplot.
SK-STAT-BOXPLOT
- I** Determine and interpret a five-number summary.
SK-STAT-FIVE-NUMBER

M39-STA-009

I can use percentiles and z-scores to locate and compare values within distributions.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Determine and interpret the percentile rank of a value.
SK-STAT-PERCENTILE
- I** Calculate and interpret a z-score.
SK-STAT-ZSCORE

I can compare distributions using their shape, center, spread, and unusual features.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Select an appropriate measure of center based on distribution shape and context.
SK-STAT-CENTER-SELECT · first introduced in Math 39
- I** Compare distributions using shape, center, spread, and unusual features.
SK-STAT-DISTRIBUTION-COMPARE
- A** Describe a distribution by modality, symmetry, and skewness.
SK-STAT-DISTRIBUTION-SHAPE · first introduced in Math 39

Probability

REQUIRED

Resource coverage: No mapped textbook section

M39-PRO-001

I can organize categorical data in a contingency table or Venn diagram.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Construct and interpret a contingency table.
SK-PROB-CONTINGENCY-TABLE
 - I** Represent categorical events and their overlap with a Venn diagram.
SK-PROB-VENN-DIAGRAM
-

M39-PRO-002

I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Calculate and interpret a conditional probability.
SK-PROB-CONDITIONAL
 - I** Calculate probability from observed relative frequency.
SK-PROB-EMPIRICAL
 - I** Calculate and interpret marginal and joint probabilities.
SK-PROB-MARGINAL-JOINT
-

M39-PRO-003

I can identify an experiment, outcomes, events, and a sample space.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Identify outcomes, events, and the sample space of an experiment.
SK-PROB-SAMPLE-SPACE
-

M39-PRO-004

I can calculate theoretical probabilities when outcomes are equally likely.

Resource: No mapped textbook section

SUPPORTING SKILLS

- A** Identify outcomes, events, and the sample space of an experiment.
SK-PROB-SAMPLE-SPACE · first introduced in Math 39
 - I** Calculate theoretical probability for equally likely outcomes.
SK-PROB-THEORETICAL
-

M39-PRO-005

I can use complements to calculate probabilities.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Use the complement rule to calculate probability.
SK-PROB-COMPLEMENT

M39-PRO-006

I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Calculate the probability of a compound and event.
SK-PROB-AND
- D** Calculate and interpret a conditional probability.
SK-PROB-CONDITIONAL · first introduced in Math 39
- I** Determine whether events are independent or dependent.
SK-PROB-INDEPENDENCE

M39-PRO-007

I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Determine whether events are overlapping or disjoint.
SK-PROB-DISJOINT
- I** Calculate the probability of a compound or event.
SK-PROB-OR

M39-PRO-008

I can compare experimental and theoretical probability and explain the law of large numbers.

Resource: No mapped textbook section

SUPPORTING SKILLS

- D** Calculate probability from observed relative frequency.
SK-PROB-EMPIRICAL · first introduced in Math 39
- I** Explain how experimental probability stabilizes with many trials.
SK-PROB-LARGE-NUMBERS
- D** Calculate theoretical probability for equally likely outcomes.
SK-PROB-THEORETICAL · first introduced in Math 39

Expected Value

EXTENSION

Resource coverage: No mapped textbook section

M39-EV-001

I can calculate the expected value of a discrete probability situation.

Resource: No mapped textbook section

SUPPORTING SKILLS

I Calculate the expected value of a discrete probability situation.

SK-PROB-EXPECTED-VALUE

M39-EV-002

I can interpret expected value to evaluate a game, risk, or long-run decision.

Resource: No mapped textbook section

SUPPORTING SKILLS

I Interpret expected value as a long-run average for decision-making.

SK-PROB-EXPECTED-INTERPRET

D Calculate the expected value of a discrete probability situation.

SK-PROB-EXPECTED-VALUE · first introduced in Math 39

Linearization of Nonlinear Data

EXTENSION

Resource coverage: No mapped textbook section

M39-POW-008

I can linearize power or exponential data and translate between the linearized and original models.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I** Transform exponential data so that a linear model can be fit to its logarithms.
SK-REG-LOG-LINEARIZE
- I** Transform power data so that a linear model can be fit on logarithmic axes.
SK-REG-LOGLOG-LINEARIZE
- I** Translate a linearized regression equation back to its original nonlinear form.
SK-REG-UNLINEARIZE